

Seven projects build up coding skills. Project pages are highlighted with a blue band

Simple step-by-step instructions guide you through each project

PYTHON

BUBBLE BLASTER 171

Distance between points

It's useful to know the distance between two points. You can use a well-known mathematical formula to calculate it out.

Get the position of the first object
Get the position of the second object
Gives back the distance between them

$$\sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

HIDE AND SEEK 43

Change the numbers in the blocks to set how strong the effect is

pixelate effect by 25

effect to 0

aphic effects

g graphic effects

hic effects in Scratch can be change a sprite's appearance (its shape, they're fun to ent with.

ou'll never know where I'll appear next!

sprite his les

ctors' block ndom position

ck random -150 to 150

This block makes the ghost move slowly, hidden from view

13 Now update the main game loop to use the functions you have just created. Remember that the order is important, so make sure you put everything in the right place. Then run the code. Bubbles should burst when they hit the sub. Check the shell window to see the score.

```
score = 0
#MAIN GAME LOOP
while True:
    if randint(1, BUB_CHANCE) == 1:
        create_bubble()
    move_bubbles()
    clean_up_bubs()
    score += collision()
    print(score)
    window.update()
    sleep(0.01)
```

This pauses the action for a very short time—try removing this and see what happens

EXPERT TIPS Python shortcut

The code "score += collision()" is a shortcut for writing "score = score + collision()". It adds the collision score to the total score, then updates the total score. Code like this is common, so a shortcut is useful. You can also do the same thing using the "--" symbol. For example, "score -= 10" is the same as "score = score - 10".



Don't forget to save your work

Each line of code is clearly labeled so you can't go wrong

This icon indicates that the project continues on the next page

Boxes give extra information: tips, definitions, and things to remember

Read on and get coding!

EXPERT TIPS

When to save

This save icon appears on the project spreads. It reminds you when to save the work you've done, so that nothing is lost if the computer crashes. Always remember to save your work frequently.



Don't forget to save your work